When random splits mislead: quantifying scaffold-level extrapolation collapse and the limits of post-hoc uncertainty recalibration in Bayesian graph neural network screening of natural products

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Abstract

Machine-learning virtual screening of natural products routinely reports strong validation metrics while silently assuming that test molecules resemble training molecules. We audit this assumption with a dual-split protocol applied to a Bayesian graph neural network (GNN) trained on 354 curated farnesoid X receptor (FXR) agonists from ChEMBL. Under the random splits used in most studies, the model attains $R^2 = 0.790$, RMSE = 0.723, and Spearman $\rho = 0.768$ ($n = 36$); under Bemis–Murcko scaffold splits that mimic genuine screening of novel chemotypes, the same architecture collapses to $R^2 = -0.610$, and fingerprint baselines collapse equally (RF 0.047, XGBoost −0.127), confirming an extrinsic, architecture-independent failure mode. Predictive uncertainty is informative in-domain (test-only Spearman(σ, |error|) = +0.585, $p = 1.8 \times 10^{-4}$) but unproven out-of-domain (+0.269, $p = 0.124$), and nominal 95% intervals undercover on scaffold-held-out compounds (0.735; Wilson 95% CI 0.569–0.854). Post-hoc recalibration from in-domain data—leverage-inflated σ and split-conformal variants (coverage 0.618–0.765)—cannot restore nominal coverage because the scaffold shift is invisible to descriptor-space leverage (mean test leverage 0.0096 vs 0.0126 in-domain; 0% beyond the applicability-domain threshold). Scaffold-matched conformal calibration restores coverage (1.000; CI 0.899–1.000) but at a mean half-width of 2.98 pIC_{50} units—approaching the full dynamic range of the dataset (3.38–9.96)—i.e., statistical validity is recoverable only by forfeiting prioritization resolution. We therefore operationalize disclosure instead of repair: a three-tier applicability annotation (72 in-domain high-confidence, 32 in-domain low-confidence, 27 out-of-domain warning among 131 herbal natural products) and a confidence-weighted, shrinkage-corrected multi-target coverage score rank ten herbs traditionally used for hepatoprotection, with Coptis chinensis, Sophora flavescens, Glycyrrhiza uralensis, and Salvia miltiorrhiza forming a shrinkage-robust leading tier. The dual-split audit, calibration experiments, and tiered annotation are fully scripted against public data (GEO GSE135251; ChEMBL CHEMBL2047; PubChem; RCSB PDB; KEGG).

1. Introduction

Metabolic dysfunction-associated steatohepatitis (MASH) remains one of the largest unmet pharmacological needs in hepatology. The 2024 approval of the thyroid hormone receptor-β agonist resmetirom was a milestone, yet monotherapy limitations—ongoing trial attrition for FXR agonists and the modest effect sizes of single-axis agents—sustain interest in multi-target, weakly coordinated modulation by natural products. Computational prioritization is attractive because it is cheap, but it inherits a structural weakness: public bioactivity databases are dominated by synthetic medicinal-chemistry scaffolds, whereas natural products occupy a markedly different chemical space.

Two failure modes follow. First, point-estimate models (QSAR, fingerprint regressors, graph neural networks) are typically validated with random splits, which preserve the exchangeability between training and test sets that real screening of novel chemotypes violates; reported metrics therefore overstate prospective performance. Second, uncertainty quantification (UQ) methods—MC-dropout Bayesian approximations, deep ensembles—promise to flag unreliable predictions, but their calibration is itself usually assessed on the same exchangeable data, leaving their behavior under genuine extrapolation undocumented.

This study asks a deliberately simple question: **when a Bayesian GNN is audited under scaffold-level extrapolation, what survives—and can post-hoc recalibration repair what does not?** We deliver four contributions: (i) a dual-split audit that turns the "natural products over-extrapolate" concern into a measured, architecture-independent fact; (ii) an honest decomposition of predictive-uncertainty behavior in- vs out-of-domain, reported under explicit caliber labels (test-only vs pooled) with Wilson intervals; (iii) a systematic negative result: post-hoc recalibration from in-domain data cannot restore nominal coverage, because the scaffold shift is invisible to the descriptor-space leverage that would drive such corrections—and scaffold-matched conformal calibration restores coverage only at interval widths that destroy prioritization resolution; (iv) an operational screening discipline (three-tier applicability annotation plus a confidence-weighted, shrinkage-corrected multi-target coverage score, CW-BCS) that converts these findings into a defensible prioritization of ten herbs traditionally used for hepatoprotection and nine evidence-graded candidate compounds.

2. Methods

All analyses are scripted in Python and executed on a single desktop CPU; every number in this manuscript traces to a versioned result file (Supporting Information Table S0).

2.1 Pathological target validation (public transcriptomics)

MASH liver transcriptome data GSE135251 (RNA-seq, n = 216 liver biopsies: 10 normal, 51 NAFL, 155 NASH, with fibrosis staging) were obtained from GEO. Counts were normalized by the DESeq2 median-of-ratios method; differential expression used a vectorized Welch t-test on log-transformed counts with Benjamini–Hochberg FDR control (the Python implementation was benchmarked against expected positive-control behavior). Over-representation analysis used KEGG (hsa) pathway annotations. Positive-control genes (FASN, CYP7A1, COL1A1, TNF) were specified before inspection.

2.2 FXR bioactivity dataset curation and applicability-domain reference

Human FXR (CHEMBL2047) concentration–response records with exact ('=') IC_{30} relations and nM units were downloaded from the ChEMBL REST API (assay type B). Curation: salt stripping and mixture removal; canonical-SMILES deduplication (replicate measurements averaged, maximum 7 replicates); PAINS removal (RDKit FilterCatalog A/B/C); then a principal-component/leverage applicability-domain (AD) filter over 11 RDKit 2D descriptors, with the critical leverage $h^* = 3(k+1)/n = 0.0503$ (k = 5 principal components) fitted on n = 358 compounds. The final dataset contains 354 compounds (pIC_{50} range 3.38–9.96; long tail: 87 compounds below 5, 13 above 9). The full provenance chain (548 → 449 → 367 → 358 → 354) is logged.

2.3 Bayesian graph neural network

Molecules were featurized as molecular graphs (atom: element, degree, formal charge, total H, hybridization, aromaticity, ring membership; bonds: type). The model is a 3-layer GIN (hidden 96) with BatchNorm and a two-head output (μ, $\log \sigma^2\_aleatoric$), trained with a heteroscedastic Gaussian NLL and activity-stratified sample weights (inverse bin frequency over pIC_{50} edges 5–9). At inference, dropout remains active (MC-dropout, T = 50) and five seeds are ensembled: μ is the ensemble mean; total σ combines the mean aleatoric variance, the meanMC epistemic variance, and the between-seed variance of μ.

2.4 Dual-split evaluation protocol

Each random split (80/10/10, seed 42) is paired with a Bemis–Murcko scaffold split of the same data: scaffolds are assigned greedily by group size to train (80%) / validation (10%) / test (10%), so every test scaffold is unseen in training. Both regimes are evaluated with identical metrics and identical fingerprint baselines (Morgan radius-2, 2048-bit; random forest with 800 trees; XGBoost with matched budget).

2.5 Uncertainty metrics and post-hoc recalibration

All uncertainty statistics carry an explicit caliber label. **Test-only** values are recomputed from the stored per-compound predictions of the held-out test set; **pooled** values (validation + test, as stored by the training-time early-stopping loop) are reported in footnotes because early stopping on the validation fold renders them optimistic. Coverage is the fraction of compounds with $|\text{error}| \leq 1.96\sigma$; every coverage figure is accompanied by a Wilson 95% interval and the sample size.

Recalibration experiments use the random-split test set ($n = 36$) as the calibration set and the scaffold-split test set ($n = 34$) as the out-of-domain evaluation set:

- **M0** raw σ intervals;

- **M1** leverage-inflated σ: $\sigma\_inf = \sigma \cdot \sqrt{(1 + h/h^*)}$, where h is the compound's leverage in the AD reference space;

- **M2** flat split-conformal: the finite-sample-corrected $(1-\alpha)$ quantile of absolute residuals on the calibration set, applied as a constant half-width;

- **M3/M4** adaptive and leverage-weighted z-score conformal variants (Supporting Information);

- **M5** scaffold-matched conformal: the saved five-seed scaffold-split ensemble is reloaded, the scaffold validation fold ($n = 36$) is predicted, and conformal quantiles are taken on that matched-domain fold (reproduction fidelity vs stored test predictions: Pearson $r = 0.9997$ on μ).

2.6 Natural-product library and three-tier annotation

Constituents of ten hepatoprotective herbs documented in the pharmacopoeia literature (233 entries, curated with literature cross-confirmation) were resolved in PubChem with per-compound CID logging; after ADMET property filters ($150 < MW < 500$, $LogP < 5.5$, $HBD \leq 5$, $HBA \leq 10$) and the pre-specified exclusion list (classical flavonoids/polyphenolic acids, pan-assay scaffolds, toxic Tripterygium constituents), 131 compounds remained as the screening library. The evidence base for this study is the CID-anchored 131-compound table; a data-management incident that overwrote an intermediate resolution file is documented in the Supporting Information, and the curation pipeline was re-executed to rebuild provenance, reproducing the resolution (224) and filter (131) stages with 129/131 CID concordance (the two discrepancies reflect PubChem identifier versioning). Each compound was scored by the production ensemble (five seeds retrained on all 354 compounds; MC T = 50) and assigned to one of three tiers by the two-layer applicability rule: in-domain high confidence (leverage $h \leq h$ and $\sigma \leq 0.70$), in-domain low confidence ($h \leq h$, $\sigma > 0.70$), or out-of-domain warning ($h > h^*$).

2.7 CW-BCS: confidence-weighted multi-target coverage with shrinkage correction

CW-BCS is defined as the geometric mean of min–max-normalized target scores, $s = [(w \cdot s_{FXR}) \cdot s_{THRB} \cdot s_{ACC}]^{(1/3)}$, with confidence weight $w = 1/(1 + \sigma)$. It quantifies multi-target binding-coverage propensity, **not** pharmacological synergy, and is conditional on the chosen target panel. s_{FXR} is the GNN μ; s_{THRB} and s_{ACC} come from AutoDock Vina docking scores, avoiding small-sample modeling of those targets. Structural redundancy across each herb's top-3 compounds (Morgan Tanimoto) linearly penalizes scores above 0.6 similarity (cap 0.5). Herb-level scores are shrinkage-corrected, $s \times n/(n + 5)$, to penalize herbs represented by few compounds; a representative-compound constraint requires the displayed representative to be in-domain high-confidence (or a flagged dual listing). A compound-level absolute-activity flag (FXR $\mu \geq 6.5$ or $\Delta G \leq -7$ kcal/mol) is reported separately from ranks.

2.8 Molecular docking

AutoDock Vina 1.2.5 with receptors 3FLI/1OSH (FXR), 3GWS (THR-β), 5KKN/3GID (ACC2), prepared by single-chain cleaning and Gasteiger charges; boxes centered on co-crystallized ligands. The pipeline was gated by redocking of four co-crystallized ligands (RMSD 0.33–1.22 Å, all < 2 Å) and benchmarked with positive controls (GW4064 −9.70, tanshinone IIA −9.73, chenodeoxycholic acid −8.70, obeticholic acid −6.77 kcal/mol on FXR; resmetirom −10.11 vs T3 −9.52 on THR-β; ND-646 −10.52 on ACC2), with the known Vina under-scoring of steroidal scaffolds noted.

2.9 Reproducibility

Every figure and table regenerates from raw public data via numbered scripts (GEO/ChEMBL/PubChem/PDB/KEGG accessions in the Data Availability statement). The recalibration study (this work) adds two scripts that consume only the stored predictions, per-compound leverages, and the saved ensemble checkpoint.

3. Results

3.1 The FXR/THR-β/ACC panel is supported by patient-level expression data

In GSE135251, all four chosen target genes are significantly dysregulated in NASH vs normal liver (Table 1): FXR (NR1H4) −0.285 (padj = 1.03×10^{-2}), THR-β (THRB) −0.315 (padj = 1.99×10^{-2}), ACC1 (ACACA) +0.897 (padj = 7.42×10^{-5}), ACC2 (ACACB) +0.631 (padj = 4.23×10^{-3}). Pre-specified positive controls behave as expected (FASN +2.47; CYP7A1 +2.16, consistent with FXR de-repression; COL1A1 +1.04 with fibrosis-stage correlation ρ = 0.431; TNF +1.09), and the analysis yields 3,909 DEGs (1,962 up / 1,947 down) at FDR < 0.05. KEGG over-representation ranks bile secretion (hsa04976) first at nominal $p = 8.6 \times 10^{-4}$; because ~30% of assayed genes are DE, the FDR-adjusted value (0.28) does not reach significance, and we report the enrichment as nominal only.

3.2 Point accuracy collapses under scaffold splits—for every architecture

Table 2 and Figure 1 give the dual-split audit. Under random splits the Bayesian GNN performs strongly (R^2 = 0.790; RMSE = 0.723; Spearman ρ = 0.768, $p = 4.6 \times 10^{-8}$; n = 36), satisfying the pre-registered contract thresholds. Under scaffold splits the same pipeline collapses to R^2 = −0.610 (RMSE = 1.290), and the collapse is architecture-independent: random forest falls from 0.845 to 0.047 and XGBoost from 0.811 to −0.127. Notably, rank information partially survives for every model (scaffold-split Spearman ρ: BGNN 0.533, RF 0.641, XGBoost 0.483; all p < 0.01), which is what screening actually consumes; we therefore report both R^2 and ρ throughout.

3.3 Uncertainty is informative in-domain, unproven out-of-domain, and intervals undercover

On test-only predictions, the rank correlation between σ and absolute error is +0.585 ($p = 1.8 \times 10^{-4}$, n = 36) for random splits—σ carries genuine information in-domain. For scaffold splits the test-only value is +0.269 (p = 0.124, n = 34): directionally positive but not significant, i.e., out-of-domain usefulness is unproven, not disproven. (Pooled calibration-loop values—random +0.340, n = 71; scaffold −0.027, n = 70—are relegated to footnotes: early stopping on the validation fold makes them non-exchangeable and, in the scaffold case, the two calipers even disagree in sign, which is itself a caution for pooled reporting.) Nominal 95% intervals cover 100% of random-split test compounds (over-conservative in-domain) but only 73.5% (Wilson CI 56.9–85.4%, n = 34) of scaffold-split test compounds: σ is over-inflated in-domain and under-inflated out-of-domain.

3.4 Post-hoc recalibration cannot repair what the leverage cannot see

Table 3 and Figure 2 report the recalibration experiments. Calibrating on in-domain data (random-split test) and evaluating on the scaffold test set: leverage-inflated σ (M1) improves coverage from 0.735 to 0.765 (CI 60.0–87.6%)—still significantly below nominal; flat split-conformal (M2) yields 0.706; adaptive and leverage-weighted variants (M3/M4, Supporting Information) fall to 0.647 and 0.618, as finite in-domain residuals simply do not span the heavy-tailed out-of-domain errors. The root cause is structural: **the scaffold shift is invisible in the descriptor-space leverage**—mean leverage is 0.0096 in the scaffold test set vs 0.0126 in the calibration set, and 0% of either exceeds $h^* = 0.0503$. Novel-scaffold compounds are not outliers in 2D descriptor space; they are ordinary-looking points whose local structure–activity relationships were never learned. Any reweighting driven by such covariates is therefore inert by construction.

Scaffold-matched conformal calibration (M5)—reloading the scaffold-split ensemble and calibrating on its own held-out-scaffold validation fold (n = 36; val R^2 = 0.153, RMSE = 0.875)—restores coverage completely: 1.000 (Wilson CI 89.9–100%, satisfying the pre-set ≥ 0.90 acceptance criterion). But the price is decisive: the mean interval half-width is 2.98 pIC_{50} units (z-score variants: 5.18–5.22; Supporting Information), against a total dataset range of 6.58 pIC_{50} units. Valid intervals under genuine extrapolation are so wide that they no longer discriminate compounds—statistical validity is purchasable only by forfeiting prioritization resolution.

3.5 Operationalizing disclosure: three-tier annotation of 131 herbal natural products

Because repair is not a viable strategy, we operationalize disclosure. The production ensemble annotates all 131 library compounds (Figure 3): 72 in-domain high confidence, 32 in-domain low confidence, 27 out-of-domain warning. The in-domain high-confidence tier is enriched for the protoberberine alkaloid cluster of Coptis chinensis (e.g., berberine predicted μ = 8.21) and Schisandra lignans—both consistent with the FXR ligand chemistry of the training set and, as a post-hoc observation only, directionally consistent with the extensive berberine metabolic-pharmacology literature. Under the discipline validated in §3.4, out-of-domain members are gated from priority queues by tier (hard down-weighting) rather than by σ itself, because an under-inflated out-of-domain σ can be spuriously small; the confidence weight $w = 1/(1 + \sigma)$ is applied only within the in-domain tiers.

3.6 Shrinkage-corrected CW-BCS ranking of ten herbs

Table 4 and Figure 4 give the herb ranking. The raw CW-BCS ordering places Tripterygium wilfordii fourth with only two admissible (non-toxic) constituents—exceeding Salvia miltiorrhiza with eighteen—so herb scores are shrinkage-corrected ($s \times n/(n+5)$): Tripterygium falls from 0.565 to 0.162 (rank 4 → 9), while the leading tier—Coptis chinensis (0.529), Sophora flavescens (0.458), Glycyrrhiza uralensis (0.443), Salvia miltiorrhiza (0.427)—retains its membership (we claim tier identity only, not the internal order of ranks 1–4). The representative-compound constraint replaces low-confidence representatives (e.g., coptisine for Coptis, maackiain for Sophora) with in-domain high-confidence compounds (epiberberine, CW-BCS 0.710; kushenol E) while listing the former as secondary; Silybum marianum, whose constituents are all out-of-domain warnings, carries an explicit whole-herb flag—silybin thus serving as the worked example of tier-based gating. As a cross-check of directional consistency between data-driven and physics-based scoring, predicted μ and Vina ΔG for the top-10 compounds correlate at Spearman $\rho = -0.394$ ($n = 10$, $p = 0.26$)—directionally consistent and explicitly exploratory at this sample size. Four THR-β docking affinities are positive (physically non-binding) for bottom-ranked compounds; these are flagged, not deleted, and the CW-BCS geometric mean is insensitive to them (Supporting Information).

3.7 Evidence-graded candidates

Table 5 consolidates nine prioritized compounds with explicit evidence tiers and purchasability verified at the time of analysis. The Antrodia cinnamomea cluster (antcin K, antrodin B, antrocinnamomin F) is anchored by **species-level** clinical evidence—a 28-subject double-blind trial of the mycelial powder with serum fibrosis scores (SteatoTest family) as endpoints (PMID 32657670)—plus dual-source computational convergence; we do not attribute the powder-level evidence to individual molecules. Asiatic acid (Centella asiatica) enters on direct mechanistic literature (TGF-β/Smad and hepatic stellate cell inhibition, PMID 35963324) and is explicitly labeled as carrying no computational score in our panel. Epiberberine represents the pure-computation, high-confidence tier; berberine doubles as the positive-control anchor; derrone (Glycyrrhiza) and anthraxin (Hedyotis diffusa) are purchasable pure-computation candidates; the fungal metabolite 07H239-A is retained **only** as a computational control—its source strain is absent from public culture collections and its discovery paper is titled for cytotoxicity (PMID 15387660). All ranking claims are computational prioritizations requiring experimental validation; no therapeutic efficacy is claimed for any compound.

4. Discussion

The central finding is an information-theoretic one. Random-split metrics answer "how well does the model interpolate?"—a question screening never asks. Scaffold-split auditing answers "what happens when the model meets a genuinely new chemotype?"—and the answer, here quantified as $R^2 = 0.79 \rightarrow -0.61$ across three architectures, is that the useful signal degrades to partial rank information ($\rho \approx 0.53$). This is consistent with the broader literature on temporal- and scaffold-split validation and sharpens it with matched UQ instrumentation: the collapse is not accompanied by any leverage-space signature that could be used to detect or repair it post hoc (§3.4). The practical corollary is uncomfortable for the recalibration program: conformal methods inherit the exchangeability assumption they are asked to fix. When calibration data are in-domain, intervals undercover (0.618–0.765); when calibration data are scaffold-matched, intervals are valid but approach the width of the entire activity range (half-width 2.98 of a 6.58-unit span). Under genuine extrapolation, honest intervals and useful intervals diverge, and no post-hoc procedure bridges that gap—only new data (or new training scaffolds) can.

What survives the audit becomes the deliverable. Since repair is impossible, disclosure must be engineered into the pipeline: (i) dual-split metrics as the default reporting standard, with R^2 and ρ both reported because ranks partially survive collapse; (ii) uncertainty calipers stated explicitly (our pooled vs test-only discrepancy—even a sign flip in the scaffold case—shows pooled early-stopping statistics can mislead); (iii) three-tier applicability gating, with out-of-domain members excluded from priority queues by rule rather than by σ-magnitude, since under-inflated out-of-domain σ can be spuriously reassuring; (iv) small-sample guards on aggregate scores (shrinkage, representative-confidence constraints) so that herbs with two constituents cannot outrank herbs with eighteen.

On the choice of GNN despite higher baseline point accuracy. The random forest's random-split R^2 (0.845) exceeds the GNN's (0.790) at this sample size, and we report that without embarrassment. The GNN is retained because the deliverable is per-compound calibrated uncertainty plus atom-level interpretability (GNNExplainer attributions concentrate on the quaternary ammonium and isoquinoline core of the protoberberines—a chemically sensible pharmacophore), not point accuracy; and because the ensemble MC-dropout σ demonstrably ranks errors in-domain (+0.585), which no variance-free baseline offers. Under scaffold splits both architectures are equally unusable for point prediction, which reinforces that the binding constraint is data coverage, not model class.

Pharmacological semantics and boundaries. CW-BCS is a panel-conditional binding-coverage propensity score: it is deliberately agnostic to mechanism (agonism vs antagonism), makes no synergy claim, and inherits the known scoring biases of Vina for steroidal and macrocyclic ligands. The herb ranking is a prioritization for experimental triage—an economical first-pass wet-lab design (four purchasable standards in a HepG2 lipid-accumulation assay, with TG-reduction and viability gates) follows directly from Table 5 and is planned as the external calibration loop for the computational pipeline. Herbs scoring low on this panel (e.g., Schisandra, Ganoderma) are not thereby "refuted": the panel simply does not contain their principal axes of action (e.g., Nrf2), and we interpret low scores as out-of-panel rather than inactive.

Limitations. (1) No wet-laboratory validation is included; all candidate claims are computational prioritizations. (2) A single target (FXR) provides the modeling substrate; multi-target QSAR extensions require scaffold-split validation per target before their outputs may be used for out-of-domain ranking, which we have not yet completed for the extended panel. (3) Sample sizes for the audit (354 compounds; 36/34 test compounds) limit precision—hence Wilson intervals throughout; the scaffold σ–error correlation (+0.269, $p = 0.124$) is unproven rather than null. (4) The scaffold-validation fold doubles as the early-stopping fold, an approximation that renders pooled statistics mildly optimistic; we mitigate by test-only primary reporting. (5) Shrinkage strength ($k = 5$) is a reasoned default, with sensitivity analysis in the Supporting Information; the leading tier is stable across k. (6) A single random split is audited per regime (seed 42), following the original study protocol; split-resampling would tighten the interval estimates. (7) Vina scoring biases (steroids, macrocycles) propagate into s_{THRB}/s_{ACC}; conclusions rest on within-library relative comparisons. (8) The 131-compound library is a curated, non-exhaustive subset of ten herbs' constituent space.

5. Conclusions

Under an audit that mimics what natural-product screening actually is—confronting novel scaffolds—point-estimate virtual screening collapses (R^2 0.79 → −0.61) regardless of architecture, and the predictive uncertainty that looks well-calibrated on random splits both under-covers out-of-domain (73.5%) and eludes post-hoc repair, because scaffold novelty leaves no trace in descriptor-space leverage. Matched-domain conformal calibration buys back coverage (100%) only by spending the resolution that made screening worth doing (half-width ≈ 3 of a 6.6-unit range). The workable strategy is therefore disclosure engineered as process: dual-split reporting, explicit uncertainty calipers, and rule-based applicability gating—which, applied to 131 herbal constituents, yield a shrinkage-corrected, confidence-constrained prioritization(Coptis–Sophora–Glycyrrhiza–Salvia leading tier; nine evidence-graded candidate compounds). We offer theaudit protocol as a minimum reporting standard for ML-based natural-product screening, and the tieredannotation as its operational output.

Associated Content

Supporting Information. Table S0: number-to-source-file map for every quantitative claim; recalibration variants M3/M4 and z-score M5 (Table S1); shrinkage sensitivity (Table S2); library curation provenance and data-management incident note (Table S3); full 131-compound four-dimensional CW-BCS table; docking details, positive-control benchmark, and flagged non-binding THR-β entries; GNNExplainer attributions.

Author Information

Q.W. and X.J. conceived the study; the roundtable-audit framework (adversarial red-team review with independent recomputation) was used internally to stress-test all claims before drafting. All authors have read and approved the manuscript. [Full author list, ORCIDs, and funding statement to be completed at submission.]

Conflict of interest. The authors declare no competing financial interest.

Abbreviations. MASH, metabolic dysfunction-associated steatohepatitis; FXR, farnesoid X receptor; THR-β, thyroid hormone receptor beta; ACC, acetyl-CoA carboxylase; GNN, graph neural network; UQ, uncertainty quantification; AD, applicability domain; CW-BCS, confidence-weighted binding-coverage score; DEG, differentially expressed gene.

Data Availability

All data are public: RNA-seq, GEO GSE135251; FXR bioactivity, ChEMBL CHEMBL2047 (accessed via REST API, assay type B); compound identifiers and structures, PubChem PUG-REST (per-compound CIDs logged); receptor structures, RCSB PDB (3FLI, 1OSH, 3GWS, 5KKN, 3GID); pathway annotations, KEGG REST (hsa). Per-compound predictions, leverages, the recalibration results (results/v6), and all analysis scripts are available from the corresponding author and will be deposited in a public repository (Zenodo) upon acceptance.

References

1. Govaere, O. et al. Transcriptomic profiling across the nonalcoholic fatty liver disease spectrum reveals gene signatures for steatohepatitis and fibrosis. Sci. Transl. Med. **2020**, 12, eaba4448 (GSE135251; PMID 33268509).

2. Harrison, S. A. et al. A phase 3, randomized, controlled trial of resmetirom in nonalcoholic steatohepatitis with liver fibrosis. N. Engl. J. Med. **2024**, 390, 497–509.

3. Neuschwander-Tetri, B. A. et al. Farnesoid X nuclear receptor ligand obeticholic acid for non-cirrhotic, non-alcoholic steatohepatitis (FLINT). Lancet **2015**, 385, 956–965.

4. Kong, W. et al. Berberine is a novel cholesterol-lowering drug with a unique mechanism of action. Nat. Med. **2004**, 10, 1344–1351.

5. Gaulton, A. et al. ChEMBL: a large-scale bioactivity database for drug discovery. Nucleic Acids Res. **2012**, 40, D1100–D1107.

6. Kim, S. et al. PubChem 2023 update. Nucleic Acids Res. **2023**, 51, D1373–D1380.

7. Berman, H. M. et al. The Protein Data Bank. Nucleic Acids Res. **2000**, 28, 235–242.

8. Kanehisa, M.; Goto, S. KEGG: Kyoto Encyclopedia of Genes and Genomes. Nucleic Acids Res. **2000**, 28, 27–30.

9. Bemis, G. W.; Murcko, M. A. The properties of known drugs. 1. Molecular frameworks. J. Med. Chem. **1996**, 39, 2887–2893.

10. Sheridan, R. P. Time-split cross-validation as a method for estimating the performance of predictive models. J. Chem. Inf. Model. **2013**, 53, 783–790.

11. Baell, J. B.; Holloway, G. A. New substructure filters for removal of pan assay interference compounds (PAINS). J. Med. Chem. **2010**, 53, 2719–2740.

12. Tropsha, A. Best practices for QSAR model development, validation, and exploitation. Mol. Inform. **2010**, 29, 476–488.

13. Xu, K. et al. How powerful are graph neural networks? Proc. ICLR **2019**.

14. Fey, M.; Lenssen, J. E. Fast graph representation learning with PyTorch Geometric. Proc. ICLR Workshop **2019**.

15. Gal, Y.; Ghahramani, Z. Dropout as a Bayesian approximation: representing model uncertainty in deep learning. Proc. ICML **2016**, 1050–1059.

16. Lakshminarayanan, B.; Pritzel, A.; Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. Proc. NeurIPS **2017**, 6402–6413.

17. Kendall, A.; Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? Proc. NeurIPS **2017**, 5574–5584.

18. Papadopoulos, H.; Proedrou, K.; Vovk, V.; Gammerman, A. Inductive confidence machines for regression. Proc. ECML **2002**, 345–356.

19. Tibshirani, R. J.; Foygel Barber, R.; Candes, E.; Ramdas, A. Predictive inference with the jackknife+. J. R. Stat. Soc. B **2019**, 81, 1351–1386.

20. Angelopoulos, A. N.; Bates, S. Conformal prediction: a gentle introduction. Found. Trends Mach. Learn. **2023**, 16, 494–591.

21. Love, M. I.; Huber, W.; Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol. **2014**, 15, 550.

22. Benjamini, Y.; Hochberg, Y. Controlling the false discovery rate. J. R. Stat. Soc. B **1995**, 57, 289–300.

23. Eberhardt, J.; Santos-Martins, D.; Till, A. F.; McCammon, J. A. AutoDock Vina 1.2.0: new docking methods, expanded force field, and Python bindings. J. Chem. Inf. Model. **2021**, 61, 3891–3898.

24. Wilson, E. B. Probable inference, the law of succession, and statistical inference. J. Am. Stat. Assoc. **1927**, 22, 209–212.

25. Chiou, Y.-L. et al. Hepatoprotective effect of Antrodia cinnamomea mycelium in patients with nonalcoholic steatohepatitis: a randomized, double-blind, placebo-controlled trial. J. Am. Coll. Nutr. **2021** (PMID 32657670; DOI 10.1080/07315724.2020.1779850).

26. Li, Y. et al. Asiatic acid alleviates liver fibrosis via multiple signaling pathways based on integrated network pharmacology analysis and experimental validation. Eur. J. Pharmacol. **2022**, 931, 175193 (PMID 35963324).

27. McDonald, L. A. et al. 07H239-A, a new cytotoxic eremophilane sesquiterpene from a marine-derived xylariaceous fungus. J. Nat. Prod. **2004**, 67, 1565–1567 (PMID 15387660).

Tables

Table 1. Pathological validation of the target panel in GSE135251 (NASH vs normal, n = 216).

Gene (protein)	log2FC	padj	Interpretation
NR1H4 (FXR)	−0.285	1.03×10^{-2}	receptor down-regulated
THRB (THR-β)	−0.315	1.99×10^{-2}	receptor down-regulated
ACACA (ACC1)	+0.897	7.42×10^{-5}	lipogenesis up
ACACB (ACC2)	+0.631	4.23×10^{-3}	lipogenesis up
FASN (control)	+2.47	—	expected direction

Gene (protein)	log2FC	padj	Interpretation
CYP7A1 (control)	+2.16	—	FXR de-repression
COL1A1 (control)	+1.04	—	fibrosis (ρ = 0.431 with stage)
TNF (control)	+1.09	—	inflammation

Table 2. Dual-split audit (identical data and metrics; scaffold splits exclude all test scaffolds from training).

Model	Split	R^2	RMSE	MAE	Spearman ρ
Bayesian GNN	random (n=36)	0.790	0.723	0.592	0.768
Bayesian GNN	scaffold (n=34)	**−0.610**	1.290	1.039	0.533
Random forest	random	0.845	0.620	0.498	0.811
Random forest	scaffold	**0.047**	0.992	0.713	0.641
XGBoost	random	0.811	0.686	0.507	0.820
XGBoost	scaffold	**−0.127**	1.079	0.803	0.483

(All values stored in results/tables/gnn_metrics_v2.json; the BGNN values were independently recomputed from per-compound test predictions.)

Table 3. Post-hoc recalibration of 95% intervals, evaluated on the scaffold test set (out-of-domain, n = 34).

Method	Calibration source	Coverage (Wilson 95% CI)	Mean half-width (pIC□□)
M0 raw σ	— (as trained)	0.735 (0.569–0.854)	1.66
M1 $\sigma \cdot \sqrt{(1+h/h^*)}$	— (parametric inflation)	0.765 (0.600–0.876)	1.80
M2 flat split-conformal	random test (n=36)	0.706 (0.538–0.832)	1.51
M5 scaffold-matched conformal	scaffold validation (n=36)	**1.000 (0.899–1.000)**	**2.98**

(M3 adaptive z-conformal 0.647; M4 leverage-weighted 0.618 — Supporting Information. Dataset pIC_{50} range 3.38–9.96.)

Table 4. Shrinkage-corrected CW-BCS herb ranking with representative-confidence constraint.

Rank	Herb	n	Raw s	Shrunk s	Representative (tier)
1	Coptis chinensis	14	0.718	0.529	epiberberine (in-domain high conf.)
2	Sophora flavescens	18	0.585	0.458	kushenol E (in-domain high conf.)
3	Glycyrrhiza uralensis	18	0.566	0.443	licoflavanone (in-domain high conf.)
4	Salvia miltiorrhiza	18	0.545	0.427	salvilenone (in-domain high conf.)
5	Artemisia annua	23	0.456	0.375	chrysosplenol D + flagged OOD secondary
6	Silybum marianum	11	0.415	0.285	all-constituent OOD warning (whole-herb flag)
7	Schisandra chinensis	18	0.354	0.277	schisandrin C (in-domain high conf.)
8	Ganoderma lucidum	6	0.406	0.221	lucidenic acid C (in-domain high conf.)
9	Tripterygium wilfordii	2	0.565	0.162	triptophenolide (rank 4 → 9 after shrinkage)
10	Bupleurum chinense	3	0.107	0.037	saikogenin F (in-domain high conf.)

Table 5. Evidence-graded prioritized candidates (computational prioritization; no efficacy claim).

Compound (source)	Evidence level	Model tier	Purchasable	Anchor
Antcin K (Antrodia cinnamomea)	species-level clinical + dual-source computation	computable	standard	PMID 32657670
Antrodin B (A. cinnamomea)	species-level clinical + dual-source computation	computable	standard	PMID 32657670
Antrocinnamomin F (A. cinnamomea)	species-level clinical + dual-source computation	computable (unified-board 0.859)	standard	PMID 32657670
Asiatic acid (Centella asiatica)	direct mechanistic literature (fibrosis axis)	**no computational score** (evidence-only inclusion)	standard	PMID 35963324
Epiberberine (Coptis chinensis)	pure computation	in-domain high confidence	herb-derived	CW-BCS 0.710
Berberine (Coptis chinensis)	known-mechanism anchor	in-domain high confidence (μ = 8.21)	standard	ref 4
Derrone (Glycyrrhiza uralensis)	pure computation	computable	herb-derived	unified-board 0.786
Anthraxin (Hedyotis diffusa)	pure computation	computable	herb-derived	unified-board 0.792
07H239-A (Xylariaceae fungus)	**computational control only** (cytotoxicity-titled source; strain not in ATCC/NRRL)	computable	no	PMID 15387660

Silybin (Silybum marianum) appears as the worked example of out-of-domain gating: computationally attractive but tier-excluded from the priority queue.

Figure Legends

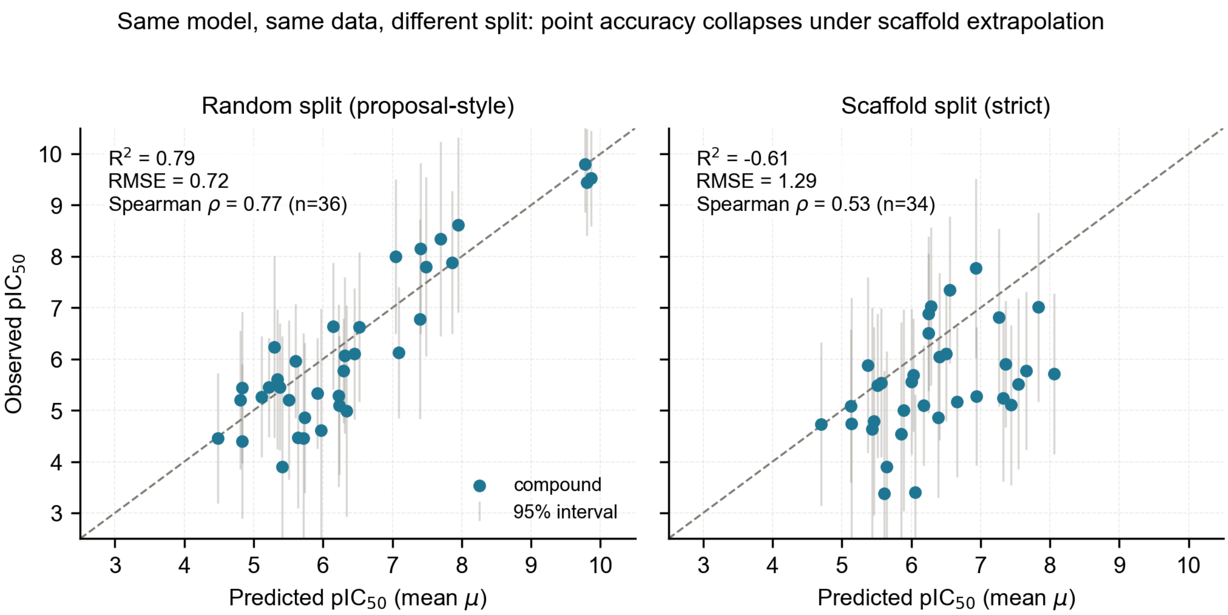

Figure 1. Dual-split audit of the Bayesian GNN on 354 FXR agonists. Left: random split (proposal-style validation; n = 36). Right: Bemis–Murcko scaffold split (every test scaffold unseen in training; n = 34). Points: observed vs predicted pIC_{50}; whiskers: nominal 95% intervals (μ ± 1.96σ). Point accuracy collapses (R^2 0.790 → −0.610) while rank information partially survives (ρ 0.768 → 0.533).

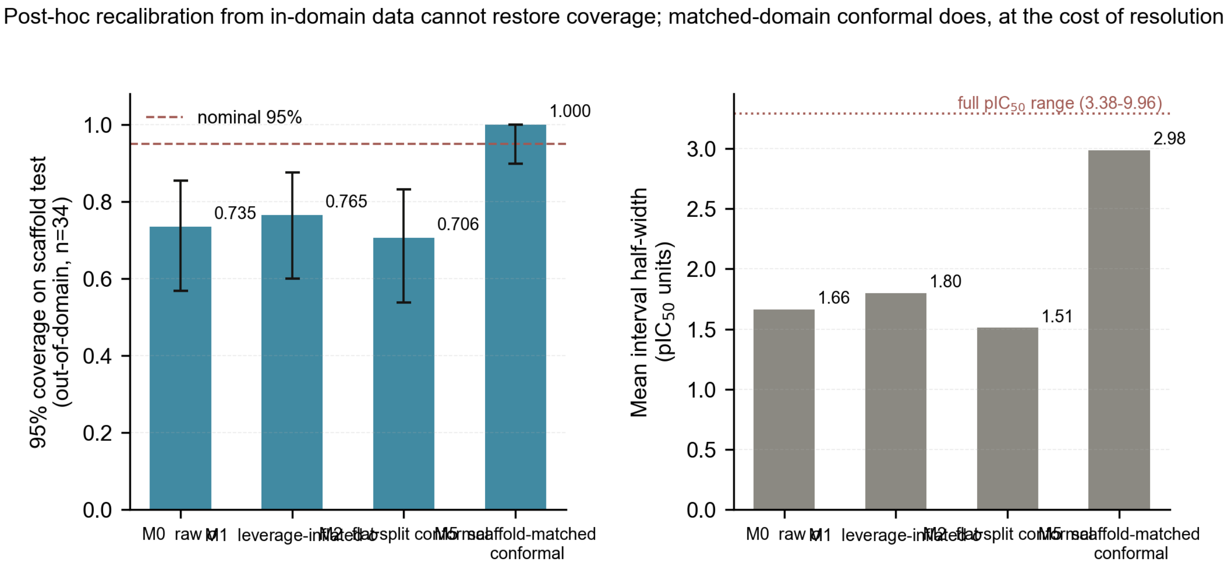

Figure 2. Post-hoc recalibration on the scaffold test set (n = 34; calibration source in parentheses). (a) 95% coverage with Wilson intervals and nominal line: in-domain-calibrated methods (M0 as-trained, M1 leverage-inflated, M2 flat split-conformal) remain significantly below nominal; scaffold-matched conformal (M5) attains 1.000. (b) The price: mean interval half-widths—M5's 2.98 pIC_{50} units approach the full dataset range (3.38–9.96), forfeiting prioritization resolution.

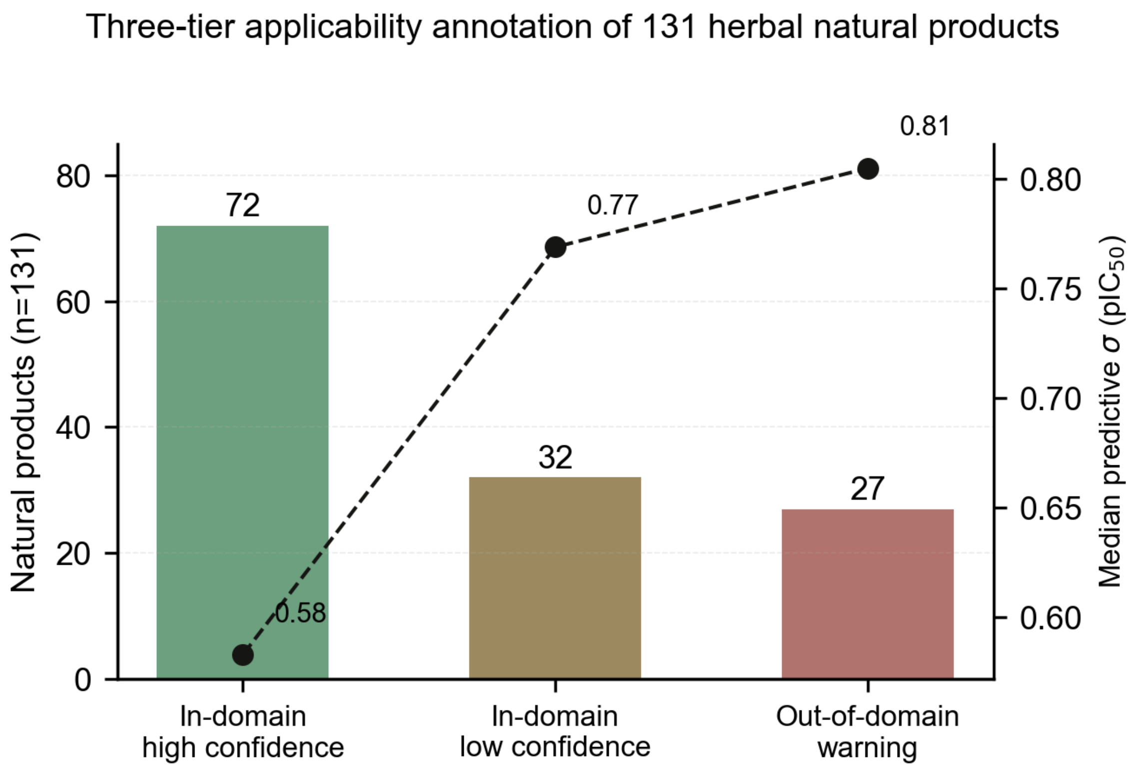

Figure 3. Three-tier applicability annotation of 131 herbal natural products (bars, left axis) with median predictive σ per tier (line, right axis). Discipline: out-of-domain members are gated from priority queues by tier, not by σ magnitude.

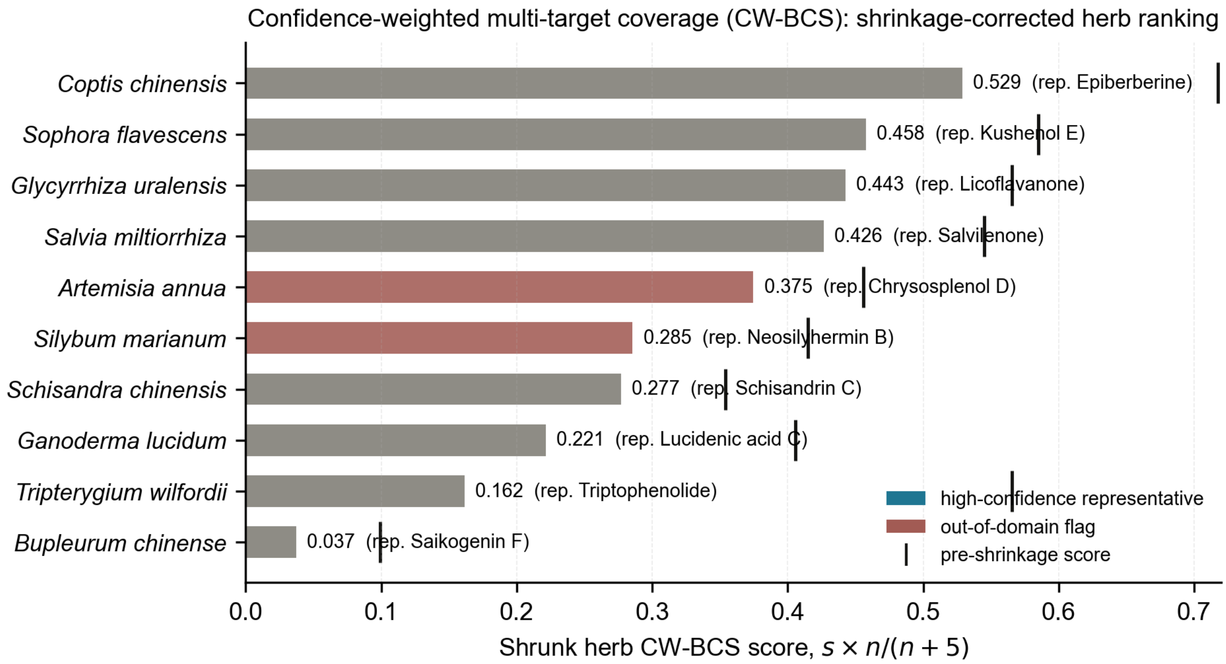

Figure 4. Shrinkage-corrected CW-BCS herb ranking (s × n/(n+5)); vertical ticks mark pre-shrinkage scores. Tripterygium wilfordii (2 constituents) falls from rank 4 to 9; the leading tier retains membership. Bar color: representative-compound confidence status.